

rate is always higher than the deposit rate. Lastly, a smart contract–enforced DAI debt ceiling can be adjusted to allow for more or less supply to meet the current level of demand. If the protocol is at the debt ceiling, no new DAI is able to be minted in new vaults until the old debt is paid or the ceiling is raised.

To stay above the liquidation threshold, a user can deposit more collateral into the vault to keep the DAI safely collateralized. When a position is deemed to be under the liquidation ratio, a keeper can initiate an auction (i.e., sell some of the ETH collateral⁶) to liquidate the position and close the vault holder's debt. The *liquidation penalty* is calculated as a percentage of the debt and is deducted from the collateral in addition to the amount needed to close the position.

After the auction, any remaining collateral reverts to the vault owner. The liquidation penalty acts as an incentive for market participants to monitor the vaults and to trigger an auction when a position becomes undercollateralized. If the collateral drops so far in value that the DAI debt cannot be fully repaid, the position is closed and the protocol accrues *protocol debt*. A buffer pool of DAI exists to cover it up to a certain amount. The solution involves the governance token MKR and the governance system.

The MKR token controls MakerDAO. Holders of the token have the right to vote on protocol upgrades, including supporting new collateral types and tweaking parameters such as collateralization ratios. MKR holders are expected to make decisions in the best financial interest of the platform.